

1. Electric current is the flow of:

- A) Voltage
- B) Charge
- C) Resistance
- D) Power

Answer: B

2. Unit of current is:

- A) Volt
- B) Watt
- C) Ampere
- D) Ohm

Answer: C

3. Voltage is defined as:

- A) Current $\times$ Resistance
- B) Work done per unit charge
- C) Power per unit time
- D) Energy per unit current

Answer: B

4. Unit of voltage is:

- A) Ampere
- B) Joule
- C) Volt
- D) Watt

Answer: C

5. Resistance opposes:

- A) Voltage
- B) Current
- C) Power
- D) Energy

Answer: B

6. Unit of resistance is:

- A) Volt
- B) Ampere
- C) Ohm
- D) Watt

Answer: C

7. Conductors have:

- A) High resistance
- B) Low resistance
- C) No resistance
- D) Infinite resistance

Answer: B

8. Insulators have:

- A) Low resistance
- B) High resistance
- C) Zero resistance
- D) No voltage

Answer: B

9. Charge is measured in:

- A) Ampere

- B) Coulomb
- C) Watt
- D) Volt

Answer: B

10. 1 Ampere equals:

- A) 1 Coulomb/sec
- B) 1 Volt/sec
- C) 1 Joule/sec
- D) 1 Watt/sec

Answer: A

11. Ohm's Law is:

- A) $V = IR$
- B) $I = VR$
- C) $R = VI$
- D) $P = VI$

Answer: A

12. Resistance increases with:

- A) Area
- B) Length
- C) Voltage
- D) Current

Answer: B

13. Resistance decreases with:

- A) Length
- B) Area
- C) Voltage
- D) Current

Answer: B

14. Resistivity unit is:

- A) Ohm
- B) Ohm-meter
- C) Volt
- D) Watt

Answer: B

15. Current is proportional to:

- A) Resistance
- B) Voltage
- C) Power
- D) Energy

Answer: B

16. Electric power formula:

- A) $P = VI$
- B) $P = IR$
- C) $P = V/R$
- D) $P = I/R$

Answer: A

17. Unit of power is:

- A) Volt
- B) Watt
- C) Joule

D) Ampere

Answer: B

18. Energy formula:

A) $V \times I$

B) Power $\times$ Time

C) $I \times R$

D) V/R

Answer: B

19. Unit of energy:

A) Watt

B) Joule

C) Volt

D) Ampere

Answer: B

20. 1 kWh equals:

A) 1000 J

B) 3600 J

C) 3.6×10^6 J

D) 36 J

Answer: C

21. In series circuit current is:

A) Same

B) Different

C) Zero

D) Infinite

Answer: A

22. In parallel circuit voltage is:

A) Same

B) Different

C) Zero

D) Infinite

Answer: A

23. Total resistance in series:

A) Sum

B) Product

C) Difference

D) Reciprocal

Answer: A

24. Total resistance in parallel:

A) Sum

B) Reciprocal sum

C) Product

D) Difference

Answer: B

25. Short circuit means:

A) High resistance

B) Zero resistance

C) Infinite resistance

D) Medium resistance

Answer: B

26. KCL is based on:

- A) Energy conservation
- B) Charge conservation
- C) Power law
- D) Voltage law

Answer: B

27. KVL is based on:

- A) Current law
- B) Charge law
- C) Energy conservation
- D) Resistance law

Answer: C

28. Sum of currents at node:

- A) Zero
- B) Infinite
- C) Maximum
- D) Minimum

Answer: A

29. Sum of voltages in loop:

- A) Zero
- B) Infinite
- C) Constant
- D) Maximum

Answer: A

30. AC stands for:

- A) Alternating Current
- B) Active Current
- C) Average Current
- D) Applied Current

Answer: A

31. Frequency unit is:

- A) Volt
- B) Hertz
- C) Ampere
- D) Watt

Answer: B

32. Standard frequency in India:

- A) 60 Hz
- B) 50 Hz
- C) 100 Hz
- D) 25 Hz

Answer: B

33. Time period is:

- A) $1/f$
- B) f
- C) f^2
- D) $1/f^2$

Answer: A

34. RMS value means:

- A) Peak
- B) Average
- C) Effective
- D) Zero

Answer: C

35. Ammeter measures:

- A) Voltage
- B) Current
- C) Power
- D) Resistance

Answer: B

36. Voltmeter measures:

- A) Current
- B) Voltage
- C) Power
- D) Energy

Answer: B

37. Wattmeter measures:

- A) Voltage
- B) Current
- C) Power
- D) Energy

Answer: C

38. Fuse protects from:

- A) Voltage
- B) High current
- C) Resistance
- D) Power

Answer: B

39. Battery converts:

- A) Electrical → Mechanical
- B) Chemical → Electrical
- C) Heat → Electrical
- D) Mechanical → Electrical

Answer: B

40. Good conductor example:

- A) Rubber
- B) Plastic
- C) Copper
- D) Wood

Answer: C

41. Insulator example:

- A) Copper
- B) Aluminium
- C) Glass
- D) Silver

Answer: C

42. Power factor is:

- A) V/I
- B) $\cos\theta$

C) I/R

D) V^2

Answer: B

43. Earthing is used for:

A) Lighting

B) Safety

C) Voltage increase

D) Current increase

Answer: B

44. Electric shock occurs due to:

A) Voltage

B) Current

C) Resistance

D) Power

Answer: B

45. Switch is used to:

A) Measure

B) Control

C) Increase

D) Decrease

Answer: B

46. Load consumes:

A) Energy

B) Voltage

C) Current

D) Resistance

Answer: A

47. Circuit is:

A) Open path

B) Closed path

C) Broken path

D) Infinite path

Answer: B

48. Open circuit has:

A) Zero resistance

B) Infinite resistance

C) Low resistance

D) Medium resistance

Answer: B

49. Power loss in resistor:

A) I^2R

B) V^2/R

C) Both

D) None

Answer: C

50. SI system stands for:

A) Standard Units

B) International System of Units

C) Scientific Units

D) System Index

Answer: B